BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT

(An Autonomous Institution affiliated to VTU, Belagavi)
Yelahanka, Bengaluru- 560 119

Python Programming		Semester	I/II
Course Code	1BPLC105B	CIE Marks	50
Teaching Hours/Week (L: T:P: S)	3:0:2:0	SEE Marks	50
Total Hours of Pedagogy	40	Total Marks	100
Credits	04	Exam Hours	03
Scheme	2025	Academic Year	2025-26
Examination type (SEE)	Theory		

Course outcome (Course Skill Set)

At the end of the course, the student will be able to:
CO1: Understand syntax and semantics of python programming
CO2: Apply knowledge of python programming for different applications.
CO3: Develop python programs to realize various computational applications
CO4: Demonstrate the conduction of experiments for the given requirement using python
CO5: Take part in self-study individually, through an online course related to python programming and complete the course successfully.

Module-1

Python Basics: Entering Expressions into the Interactive Shell, The Integer, Floating-Point, and String Data Types, String Concatenation and Replication, Storing Values in Variables, Your First Program, Dissecting Your Program

Flow control: Boolean Values, Comparison Operators, Boolean Operators, Mixing Boolean and Comparison Operators, Elements of Flow Control, Program Execution, Flow Control Statements, Importing Modules, Ending a Program Early with sys. exit ()

Textbook 1: Chapters 1 – 3 Number of Hours: 08

Module-2

Functions: def Statements with Parameters, Return Values and return Statements, The None Value, Keyword Arguments, and print (), Local and Global Scope, the global Statement, Exception Handling, A Short Program: Guess the Number

Classes and objects: Programmer-defined types, Attributes, Rectangles, Instances as return values, Objects are mutable, Copying

Classes and functions: Time, Pure functions, Modifiers

Textbook 1: Chapter 4

Textbook 2: Chapters 16, 14 Number of Hours:08

Module-3

Classes and methods: Object-oriented features, Printing objects, Another example, A more complicated example, The init method, The str method, Operator overloading, Type-based dispatch, Polymorphism,

Lists: The List Data Type, Working with Lists, Augmented Assignment Operators, Methods, Example Program: Magic 8 Ball with a List, List-like Types: Strings and Tuples, References

Textbook 2: Chapters 15

Textbook 1: Chapter 6 Number of Hours: 08

Module-4

Dictionaries and Structuring Data: The Dictionary Data Type, Pretty Printing, Using Data Structures to Model Real-World Things

Manipulating Strings: Working with Strings, Putting Strings Inside Other Strings, Useful String Methods, Numeric Values of Characters with the ord() and chr () Functions, Coping and Pasting Strings with the pyperclip Module, Project: Multi-Clipboard Automatic Messages

Textbook 1: Chapters 7-8 Number of Hours:08

Module-5

Reading and Writing Files: Files and File Paths, The File Reading/Writing Process, Saving Variables with the shelve Module, Saving Variables with the pprint. pformat () Function

Organizing Files: The shutil Module, walking a Directory Tree, Compressing Files with the zip file Module

Working with Excel Spreadsheets: Excel Documents, Installing the openpyxl Module, Reading Excel Documents

Textbook 1: Chapters 10-11,14 Number of Hours:08

PRACTICAL COMPONENTS OF IPCC

PART – A: CONVENTIONAL EXPERIMENTS

1. You are working as a software developer for "CoffeeKutira Restaurant". The restaurant manager has asked you to create a calculator program to help staff quickly calculate various bill-related operations for a single item during busy hours. The program should allow staff to choose from 4 operations:
    a) Add - Calculate total bill (food cost + tax)
    b) Subtract - Calculate discount amount (original bill - discounted bill)
    c) Multiply - Calculate total for multiple identical orders
    d) Divide - Split bill among customers
2. You're working as a cashier at a shopping mall. Write a program that asks customers how many items they want to buy. For each item, input the price, Calculate total bill and apply 10% discount if total > Rs. 100. Display itemized receipt
3. You're managing an ice cream parlour. Write a program that shows the menu, takes customer orders,
   Asks "Do you want to order more?" after each order and calculates total bill with 8% tax
4. Develop a program that uses class Student which prompts the user to enter marks in three subjects and calculates total marks, percentage and displays the score card details. [Hint: Use list to store the marks in three subjects and total marks. Use __init__() method to initialize name, USN and the lists to store marks and total, Use getMarks() method to read marks into the list, and display () method to display the score card details.]
5. Amazon.com monitors their website response times during festival sales. They collect response times (in milliseconds) for 10 random page loads during peak traffic hours.
   Response Times (ms): [245, 312, 189, 276, 298, 234, 389, 267, 301, 223]
   Compute the Average response time (Mean) and inconsistent performance (Standard Deviation) of the website.
6. A digital library wants to analyze book content to help readers find similar books and improve their recommendation system. Develop a program that can find the 10 most frequent words in each book and generate a report showing Top 10 words in a book. [Hint: Use dictionary with distinct words and their frequency of occurrences. Sort the dictionary in the reverse order of frequency and display dictionary slice of first 10 items]
7. Develop a function named DivExp which takes TWO parameters a, b and returns a value c (c=a/b). Write a suitable assertion for a>0 in function DivExp and raise an exception for when b=0. Develop a suitable program which reads two values from the console and calls a function DivExp.
8. You are developing multiplicationTable.py for a primary school teacher who wants to generate a multiplication table for numbers up to N. Develop a program that creates an NxN table taking the value N through command line arguments.
9. Consider a studData.xlsx file. The file has the USN, Name and CGPA of the students in the class. Develop a program to find the first topper of the class.

PART – B: OPEN ENDED EXPERIMENTS

Open-ended experiments are a type of laboratory activity where the outcome is not predetermined and students are given the freedom to explore, design, and conduct the experiment based on the problem statements as per the concepts defined by the course coordinator. It encourages creativity, critical thinking, and inquiry-based learning.

1. Decision Making, Looping
2. Functions
3. Classes and Objects
4. Files

Suggested Learning Resources: (Textbook/ Reference Book/ Manuals):

Text books:

l. Sweigart, "Automate the Boring Stuff with Python Practical Programming for total beginners", 3rd Edition, No Starch Press, 2025.

2. Allen B. Downey, "Think Python: How to Think Like a Computer Scientist", 3rd Edition, Oreilly, Green Tea Press, 2024.

Reference books / Manuals:

- Introduction to Computer Science Using Python: A Computational Problem-Solving Focus, by Charles Dierbach, Wiley India Edition, 2018, ISBN: 978-81-265-5601-4
- PYTHON PROGRAMMING AN INTRODUCTION TO COMPUTER SCIENCE, John Zelle, Franklin, Beedle & Associates Inc,2016,ISBN 9781590282755
- Practical Programming, Third Edition An Introduction to Computer Science Using Python 3.6 by Paul Gries, Jennifer Campbell, Jason Montojo,O'Reilly; 3rd edition (9 January 2018)

Web links and Video Lectures (e-Resources):

- **https://www.learnbyexample.org/python/**
- **https://www.learnpython.org/**
- **https://pythontutor.com/visualize.html#mode=edit**

Teaching-Learning Process (Innovative Delivery Methods):

The following are sample strategies that educators may adopt to enhance the effectiveness of the teaching-learning process and facilitate the achievement of course outcomes.

1. Python Tutor
2. Online Coding Platforms (hacker earth, Hacker rank, Leetcode etc.)

Assessment Structure:

The assessment for each course is equally divided between Continuous Internal Evaluation (CIE) and the Semester End Examination (SEE), with each component carrying **50% weightage** (i.e., 50 marks each). The CIE component consists of 25% weightage for theory and 25% for lab. Theory components have two IA tests and one Continuous Comprehensive Assessments (CCA). Each IA is for 40Marks. 80Marks of IA to be scaled to down to 15Marks. CCA to be conducted for 10Marks. Lab components consist of continuous weekly evaluation and one practical lab exam. Continuous weekly evaluation to be done for 30marks. Average of all continuous evaluation to be scaled down to 15Marks. Practical lab internal for 50Marks to be conducted after completing all lab programs. 50 Marks to be scaled down to 10 Marks.

- To pass the **CIE**, a student must secure **a minimum of 40% of 50 marks**, i.e., **20 marks**.
- To pass the **SEE**, a student must secure **a minimum of 35% of 50 marks**, i.e., **18 marks**.
- A student is deemed to have **successfully completed the course** if the **combined total of CIE and SEE is at least 40 out of 100 marks**.

Continuous Comprehensive Assessments (CCA):

CCA will be conducted for 10 marks. It is recommended to include learning activities aimed at enhancing the holistic development of students. These activities should align with course objectives and promote higher-order thinking and application-based learning.

Learning Activity -1: MOOC Course (Marks- 10)

CIE Practical component: The CIE marks awarded in the case of the Practical component shall be based on the continuous evaluation of the laboratory report using a defined set of rubrics. Each experiment report can be evaluated for 30 marks. The summation of all the experiments marks to be scaled down to 15 marks. The laboratory test (duration 03 hours) at the end of the last week of the semester /after completion of all the experiments (whichever is early) shall be conducted for 50 marks and scaled down to 10 marks. For laboratory test and SEE, the student is required to conduct one experiment each from both Part A and Part B. Part A is assessed with a weightage of 70%, while Part B carries a weightage of 30%
Rubrics for Learning Activity (Based on the nature of learning activity, design the rubrics for each activity):

Learning Activity -1 : MOOC Course.

Rubrics to be framed as per requirement